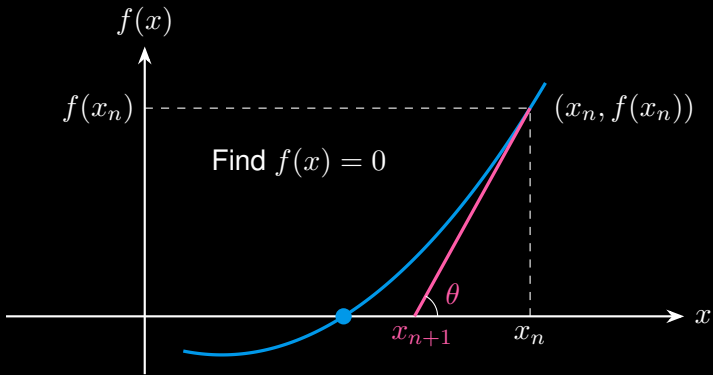


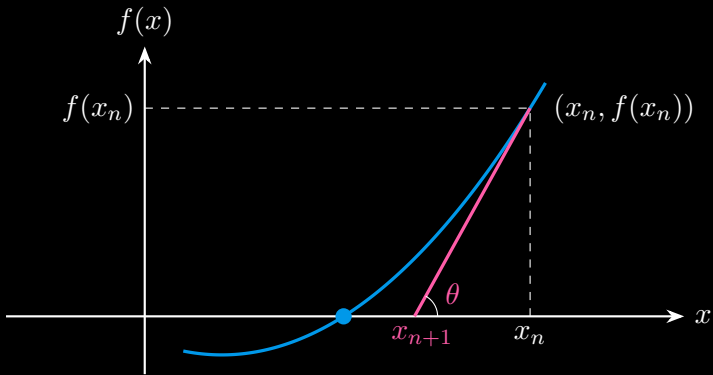
Intuitive Understanding of Newton's Method

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{0 - f(x_n)}{x_{n+1} - x_n} = f'(x_n)$$



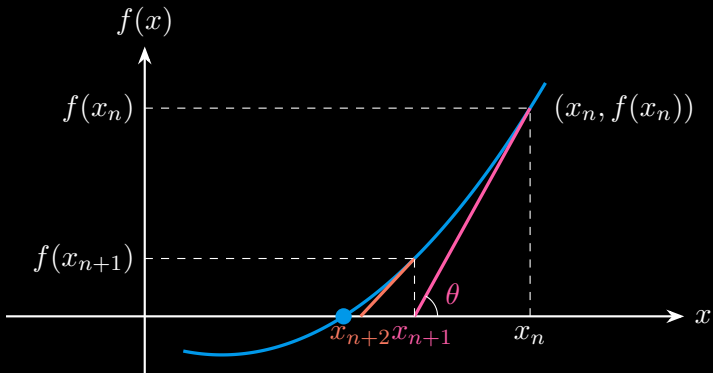
Intuitive Understanding of Newton's Method

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \quad \Rightarrow \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



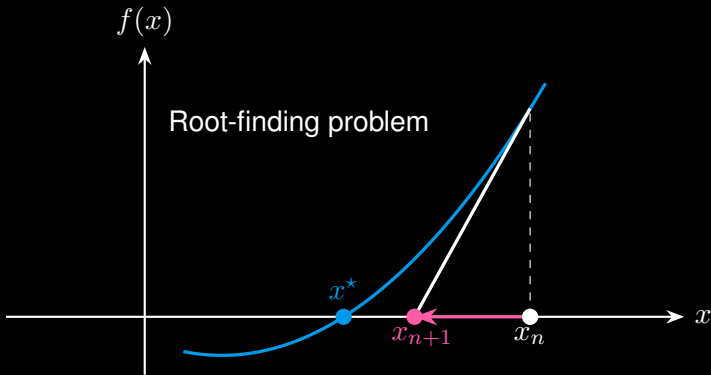
Intuitive Understanding of Newton's Method

$$x_{n+2} = x_{n+1} - \frac{f(x_{n+1})}{f'(x_{n+1})}$$



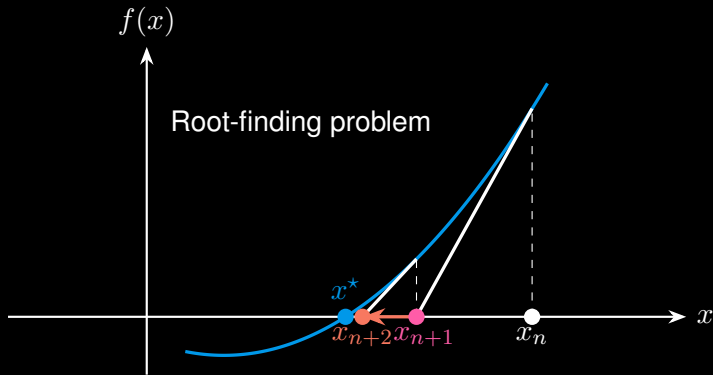
Intuitive Understanding of Newton's Method

$$x := x - \frac{f(x)}{f'(x)} \quad (\text{update } x \text{ until convergence})$$



Intuitive Understanding of Newton's Method

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Unconstrained Nonlinear Optimization

Optimization problem:

$$\min_x \underbrace{f(x)}_{\text{twice-differentiable}} \quad \Rightarrow \quad \text{Find } f'(x) = 0$$

Unconstrained Nonlinear Optimization

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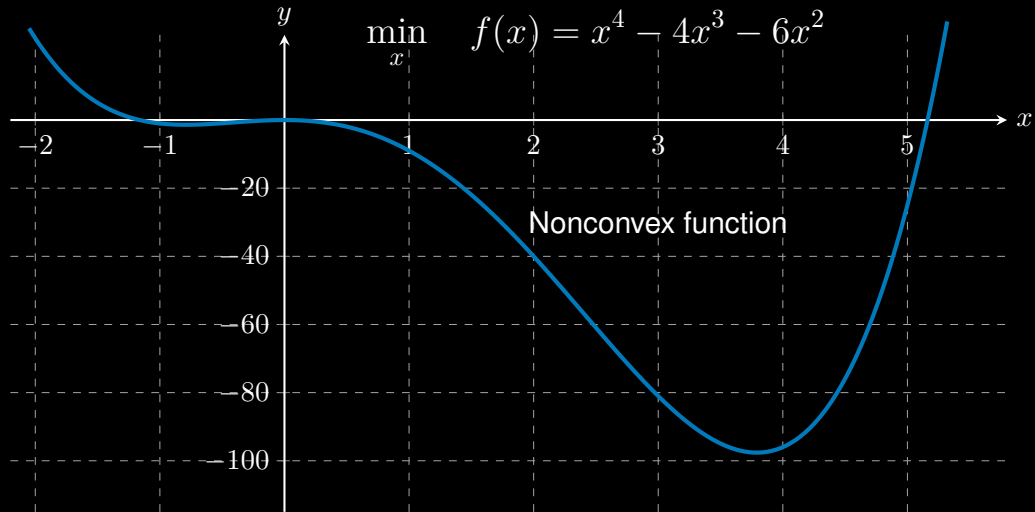
Newton's method in optimization is an iterative algorithm to find stationary points such as (local) **minimum** of a twice-differentiable function $f(x)$:

$$x := x - \frac{f'(x)}{f''(x)} \quad (\text{update } x \text{ until convergence})$$

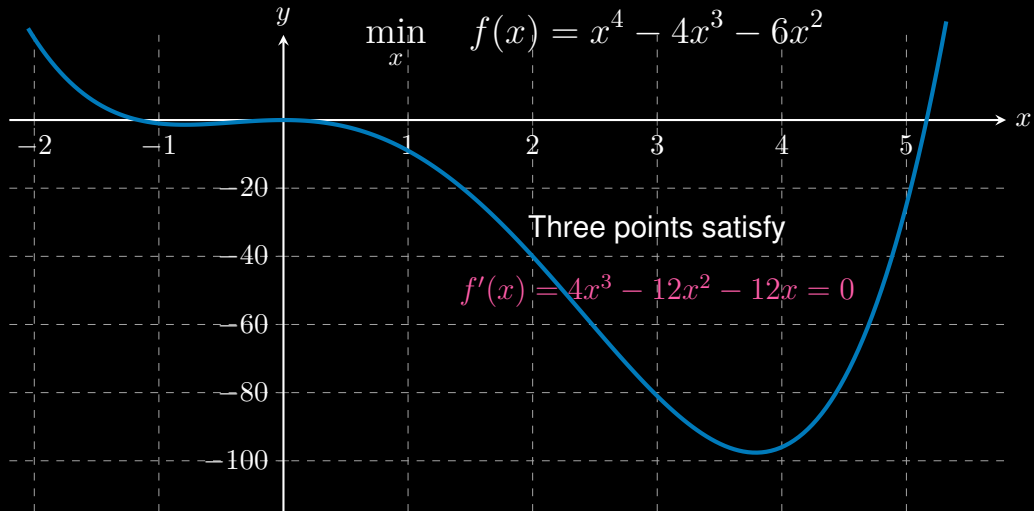
with

$f'(x)$	first-order derivative
$f''(x)$	second-order derivative

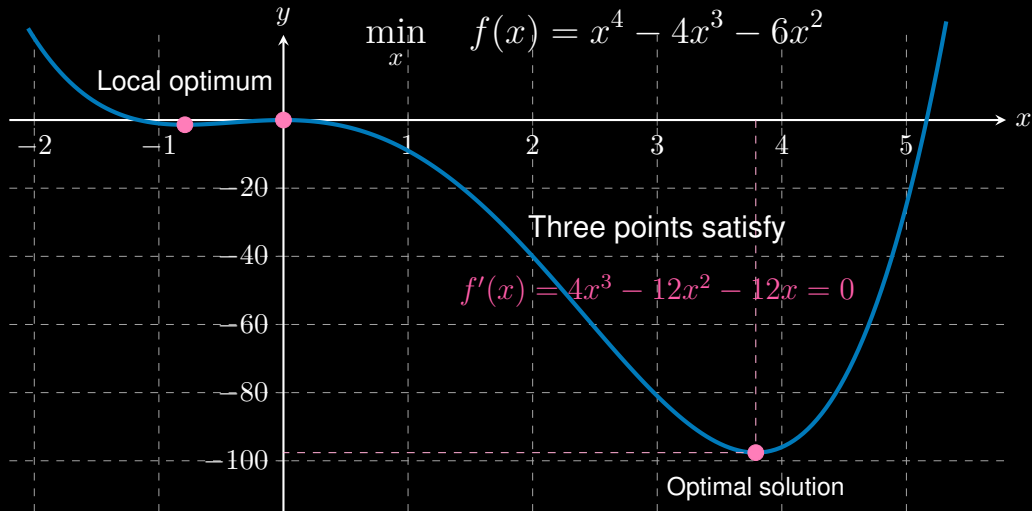
Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization



Newton's Method in Unconstrained Nonlinear Optimization

Optimization problem:

$$\min_x f(x) = x^4 - 4x^3 - 6x^2$$

Newton's method:

$$x := x - \frac{f'(x)}{f''(x)} \quad (\text{update } x \text{ until convergence})$$

with

$$f'(x) = 4x^3 - 12x^2 - 12x \quad \text{and} \quad f''(x) = 12x^2 - 24x - 12$$

Newton's Method in Unconstrained Nonlinear Optimization

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$$\min_x f(x) = x^4 - 4x^3 - 6x^2$$

Newton's method:

$$x := x - \frac{f'(x)}{f''(x)} \quad (\text{update } x \text{ until convergence})$$

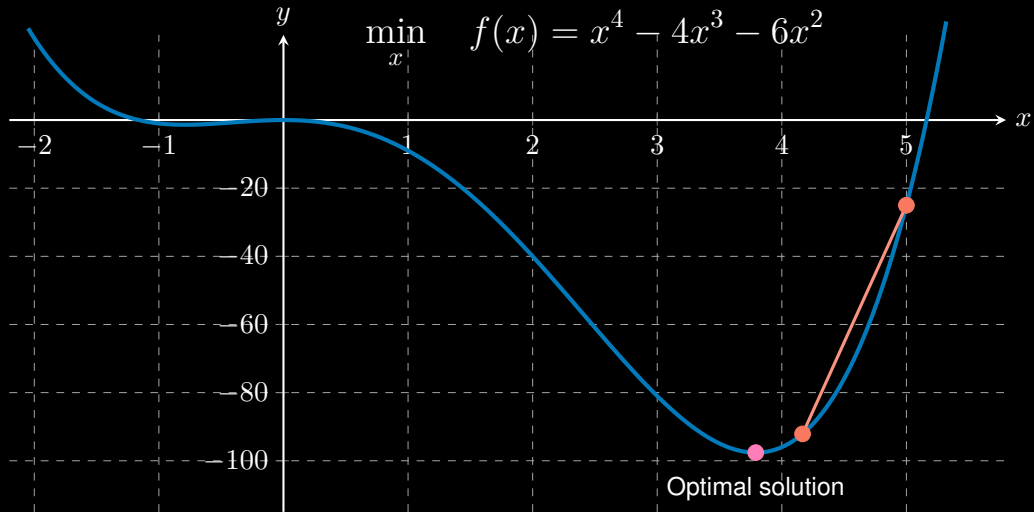
with

$$f'(x) = 4x^3 - 12x^2 - 12x \quad \text{and} \quad f''(x) = 12x^2 - 24x - 12$$

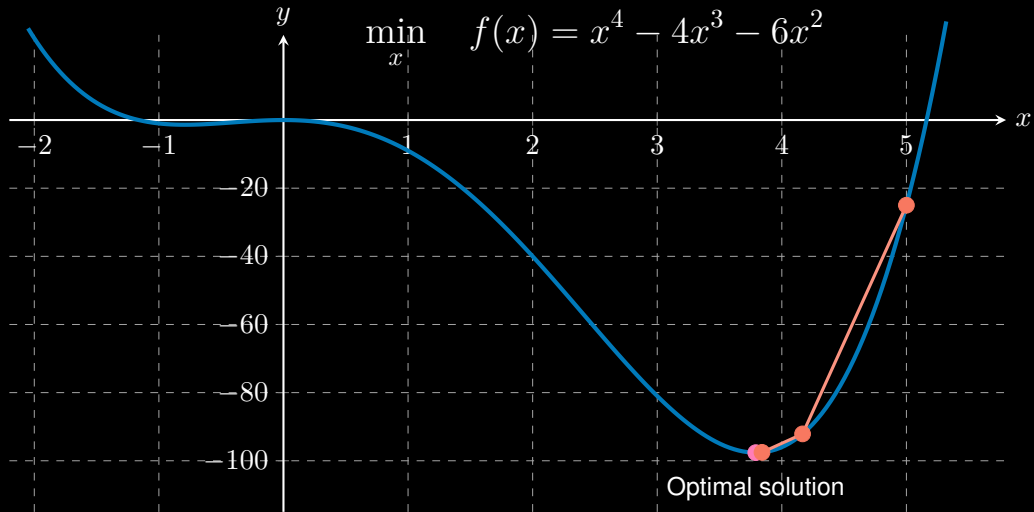
Given a starting point $x_0 = 5$:

$$f'(5) = 140, \quad f''(5) = 168 \quad \Rightarrow \quad x := 5 - \frac{f'(5)}{f''(5)} = 4.1667$$

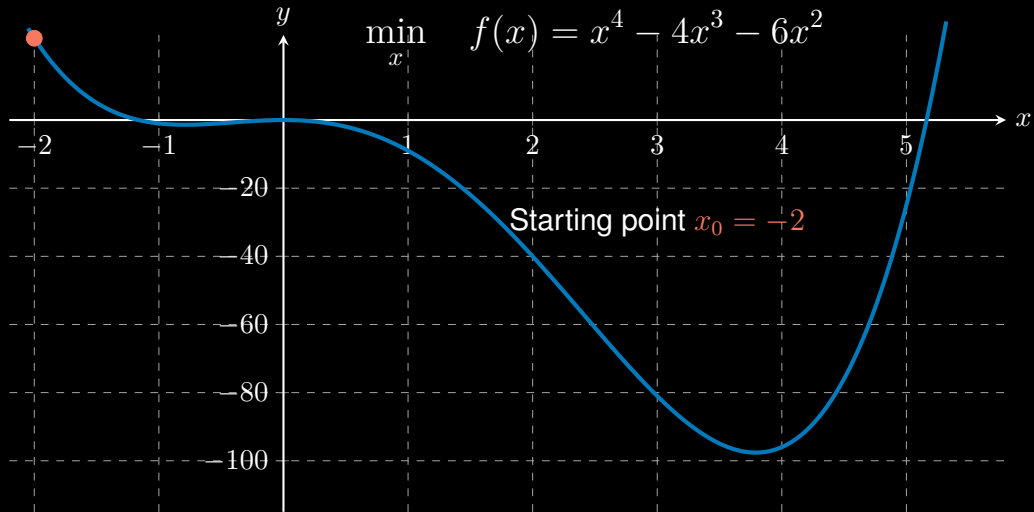
Newton's Method in Unconstrained Nonlinear Optimization



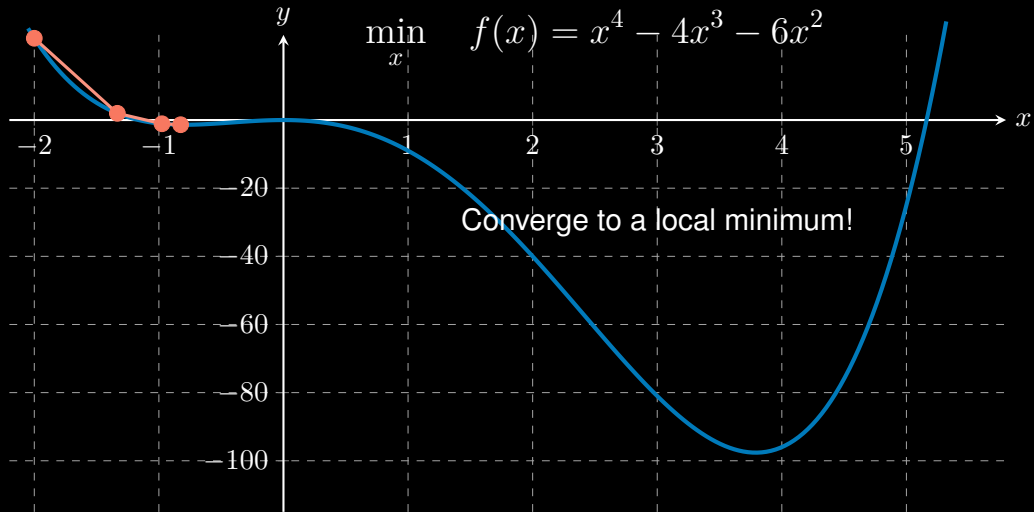
Newton's Method in Unconstrained Nonlinear Optimization



Newton's Method in Unconstrained Nonlinear Optimization



Newton's Method in Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization

Bivariate **Rosenbrock function**:

$$\min_{x_1, x_2} f(x_1, x_2) = (a - x_1)^2 + b(x_2 - x_1^2)^2$$

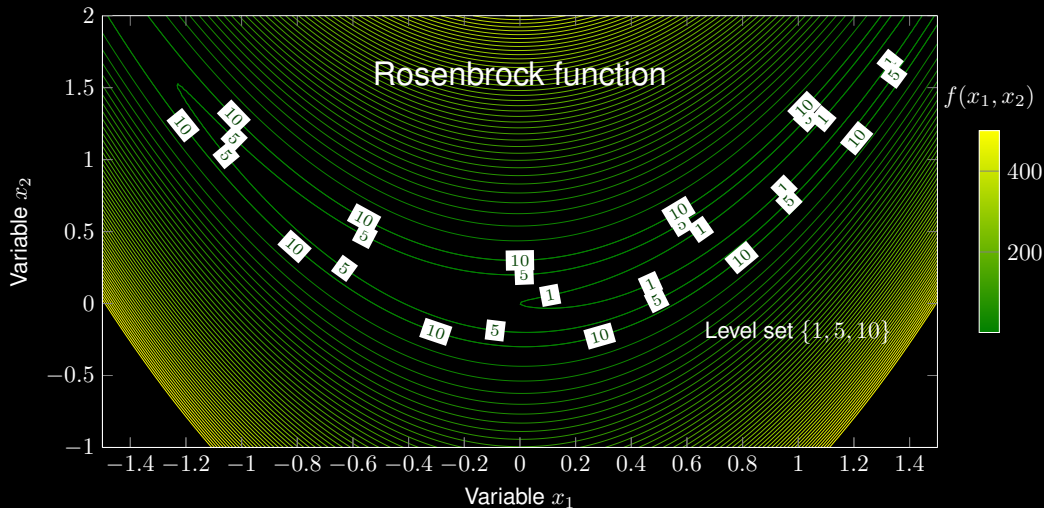
with $a = 1$ and $b = 100$.

Rosenbrock function in mathematical optimization:

- It is a **nonconvex function** introduced by *Howard H. Rosenbrock* in 1960.
- It is a commonly-used **test function for optimization**.
- It is a popular **benchmark** in statistics and machine learning.

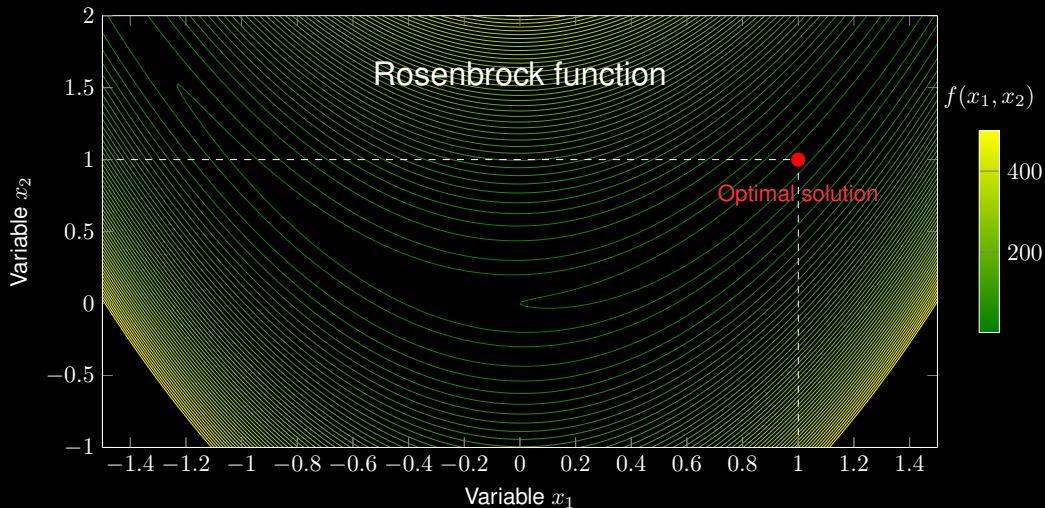
Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



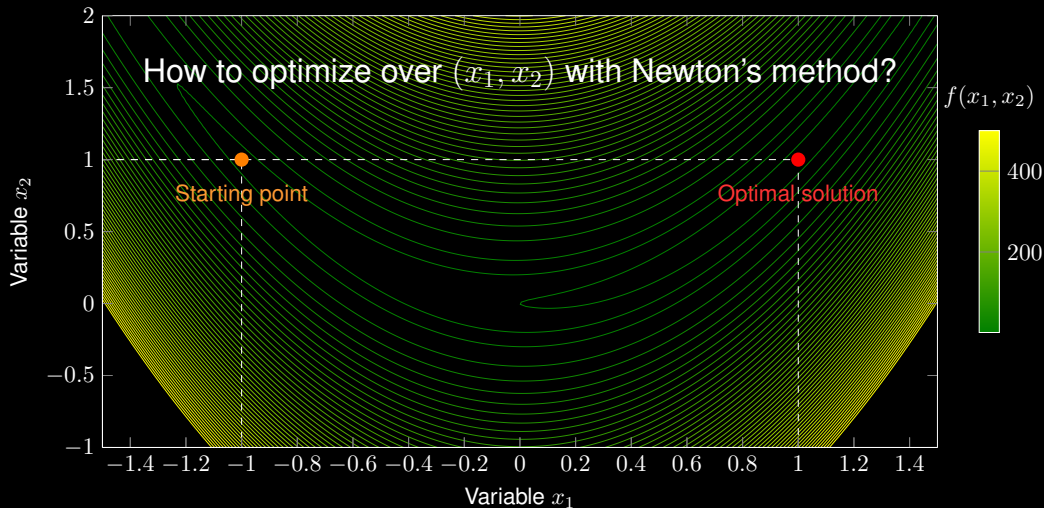
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Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



First- and Second-Order Derivatives

Optimization problem:

$$\min_{x_1, x_2} \underbrace{f(x_1, x_2)}_{\text{twice-differentiable}} = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$

First-order derivatives:

$$\frac{\partial f}{\partial x_1} = -2 + 2x_1 - 400x_1x_2 + 400x_1^3 \quad \text{and} \quad \frac{\partial f}{\partial x_2} = 200x_2 - 200x_1^2$$

First- and Second-Order Derivatives

Optimization problem:

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First-order derivatives:

$$\frac{\partial f}{\partial x_1} = -2 + 2x_1 - 400x_1x_2 + 400x_1^3 \quad \text{and} \quad \frac{\partial f}{\partial x_2} = 200x_2 - 200x_1^2$$

Second-order derivatives:

$$\frac{\partial^2 f}{\partial x_1^2} = \frac{\partial}{\partial x_1} (-2 + 2x_1 - 400x_1x_2 + 400x_1^3) = 2 - 400x_2 + 1200x_1^2$$

$$\frac{\partial^2 f}{\partial x_2^2} = \frac{\partial}{\partial x_2} (200x_2 - 200x_1^2) = 200$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = -400x_1$$

Newton's Method in Unconstrained Nonlinear Optimization

The optimization can be converted into a root finding problem:

$$\min_{x_1, x_2} f(x_1, x_2) \quad \Rightarrow \quad \text{Find} \quad \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \mathbf{0} \quad (\text{first-order derivatives} = 0)$$

Newton's method for root finding:

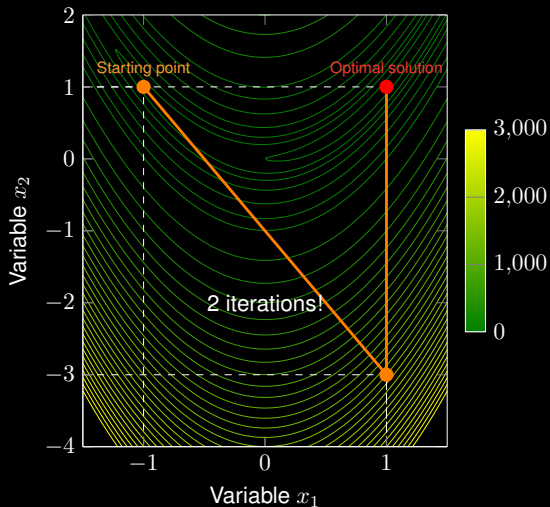
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \mathbf{H}^{-1} \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}$$

with Hessian matrix:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 - 400x_2 + 1200x_1^2 & -400x_1 \\ -400x_1 & 200 \end{bmatrix}$$

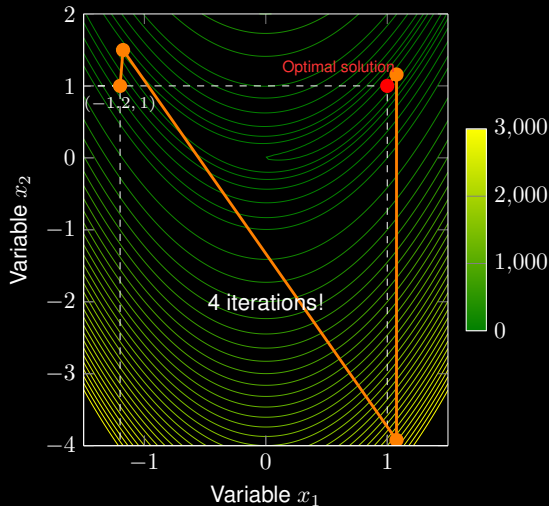
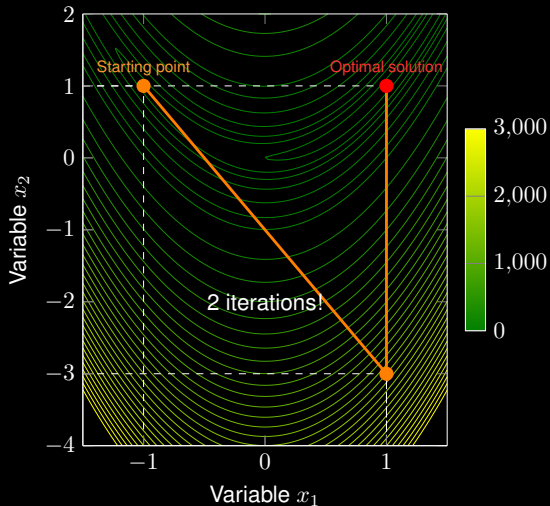
Newton's Method in Unconstrained Nonlinear Optimization

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Newton's Method in Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



Thanks for your attention!

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